

Activity 10 Instruction Manual

Activity 10 | Topic 01 | C013 | Version v1.0

Purpose

You will apply the four estimating techniques to the same work, compare what each produces, and assemble a cost baseline with contingency and management reserves.

Files used

- activity-guide.md and activity-guide.pdf - brief and source instructions
- prompts.pdf - reusable prompt and verification checklist
- mock-data.xlsx - safe fictional project data for practice

Expected output

A cost baseline built bottom-up from your WBS, with reserves and a documented estimating basis.

Before you begin

1. Form the required group and assign one person to capture evidence.
2. Open mock-data.xlsx and review the title, note and column headings.
3. Open prompts.pdf beside your chosen GenAI tool. Do not upload confidential workplace data.

Detailed procedure

1. Using the WBS from Activity 8, prompt GenAI: 'Estimate the cost of each work package in this WBS using bottom-up estimating. Show your assumed rates and quantities separately from the totals.'
2. Ask for the same total three more ways: analogous ('estimate from a similar past project'), parametric ('use a rate per unit') and three-point ('give optimistic, most likely and pessimistic').
3. Compute the three-point estimate yourself using $C_e = (C_o + 4C_m + C_p) / 6$ and compare all four figures. Explain to your group why they differ and which you would defend to a sponsor.
4. Challenge the assumed rates. Prompt: 'Which of these rates did you assume rather than know? Mark each with [ASSUMPTION].' Replace them with rates from your own organisation.
5. Roll the work package costs up to a project total - this is your cost baseline before reserves.
6. Add a contingency reserve for the identified risks you will quantify in Topic 3, and a management reserve for unforeseen in-scope work. Prompt GenAI to suggest a percentage and to justify it, then decide your own figure.
7. Confirm you can state the difference: contingency reserve is inside the cost baseline for known risks; management reserve sits outside it for unknown work.
8. Share your baseline and the reserve percentages you chose, with your justification.

Use the supplied mock data

Copy only the rows needed for the current step and tell the model the data is fictional. If the model changes an identifier or number, reject the change and rerun with an explicit preserve-values instruction.

Quality control

- You have one cost baseline built bottom-up, four comparable estimates for the same work, no unmarked assumed rates, and a stated contingency and management reserve you can justify.
- Compare the first AI draft with the final human-corrected artifact and retain both as evidence.
- Label assumptions, decisions, owners and dates so another project manager can audit the result.

Troubleshooting

If the response is generic, add the project objective, constraints, audience and required columns. If arithmetic is inconsistent, request formulas and substitutions, then calculate independently. If a source cannot be opened, remove the claim or replace it with verified evidence.

Submission evidence

Submit the completed project artifact, the final prompt, a short critique of the AI output, and any screenshot or exported file required by the activity.